```
> # PROOF OF IDENTITIES FOR a_5(n)
  with(qseries) :
  with(ETA) :
  t := gp2etaprod([7, 4, 1, -4]) :
  p0 := gp2etaprod([14, 4, 1, 4, 7, -4, 2, -4]) :
  p1 := (gp2etaprod([14, 1, 1, 7, 7, -1, 2, -7]) - 1) / 7 :
  A5 := gp2etaprod([2, 4, 49, 5, 1, -5, 98, -4]) :
  B5 := 1 :

> # GROUP 1, IDENTITY 1: U_A5(p0)
  EP5_1_1 := normal(A5 * p0) :
  RHS5_1_1 := expand(p0 * (7^4 * t^3 + 15 * 7^2 * t^2 + 8 * 7 * t) + p1 * (-7^3 * t^2 - 2 * 7^2
      * t - 7) + 7^3 * t^2 + 2 * 7^2 * t + 7) :
  proof5_1_1 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_1, 7, RHS5_1_1, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof5_1_1 := [1, -9, O(q^{37})] \quad (1)$$

```
> # GROUP 1, IDENTITY 2: U_A5(p0*t^(-1))
  EP5_1_2 := normal(A5 * p0 / t) :
  RHS5_1_2 := expand(p0 * (-7^3 * t^2 - 8 * 7 * t) + p1 * (7^2 * t + 7) - 7^2 * t - 7) :
  proof5_1_2 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_2, 7, RHS5_1_2, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$O(q^{34})$$

(2)

$$proof5_1_2 := [1, -6, O(q^{34})] \quad (2)$$

```
> # GROUP 1, IDENTITY 3: U_A5(p0*t^(-2))
  EP5_1_3 := normal(A5*p0/t^2):
  RHS5_1_3 := expand(p0*(7^4*t^2+9*7^2*t+8)+p1*(-7^3*t-8*7-t^(-1))
      +7^3*t+8*7+t^(-1)):
  proof5_1_3 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_3,7,RHS5_1_3,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 10    This means checking up to q^(11).
We find that LHS - RHS is
```

$$O(q^{38})$$

$$proof5_1_3 := [1, -10, O(q^{38})] \quad (3)$$

```
> # GROUP 1, IDENTITY 4: U_A5(p0*t^(-3))
  EP5_1_4 := normal(A5*p0/t^3):
  RHS5_1_4 := expand(p0*(7^7*t^4+19*7^5*t^3+11*7^4*t^2-165*7*t-88)+p1
      *(-7^6*t^3-18*7^4*t^2-66*7^2*t+22*7+11*t^(-1))+7^6*t^3+18*7
      ^4*t^2+66*7^2*t-22*7-11*t^(-1)):
  proof5_1_4 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_4,7,RHS5_1_4,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 14    This means checking up to q^(15).
We find that LHS - RHS is
```

$$O(q^{42})$$

$$proof5_1_4 := [1, -14, O(q^{42})] \quad (4)$$

```
> # GROUP 1, IDENTITY 5: U_A5(p0*t^(-4))
  EP5_1_5 := normal(A5*p0/t^4):
  RHS5_1_5 := expand(p0*(7^9*t^5+8*7^7*t^4-90*7^5*t^3-860*7^3*t^2-10*7
      ^3*t+720)+p1*(-7^8*t^4-7^7*t^3+90*7^4*t^2+110*7^3*t+50*7-90
```

```
      *t^(-1)) + 7^8*t^4 + 7^7*t^3-90*7^4*t^2-110*7^3*t-50*7 + 90*t^(-1))
      :
   proof5_1_5 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_5, 7, RHS5_1_5, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
We find that LHS - RHS is
```

$$O(q^{45})$$

$$proof5_1_5 := [1, -17, O(q^{45})] \quad \textbf{(5)}$$

```
> # GROUP 1, IDENTITY 6: U_A5(p0*t^(-5))
   EP5_1_6 := normal(A5*p0/t^5) :
   RHS5_1_6 := expand(p0*(-7^11*t^6-46*7^9*t^5-570*7^7*t^4-323*7^6*t^3
       + 57*7^4*t^2 + 1045*7^2*t-5016) + p1*(7^10*t^5 + 45*7^8*t^4 + 76*7^7*t
       ^3 + 285*7^5*t^2-76*7^3*t-836*7 + 627*t^(-1))-7^10*t^5-45*7^8*t^4
       -76*7^7*t^3-285*7^5*t^2 + 76*7^3*t + 836*7-627*t^(-1)) :
   proof5_1_6 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_6, 7, RHS5_1_6, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$O(q^{48})$$

$$proof5_1_6 := [1, -20, O(q^{48})] \quad \textbf{(6)}$$

```
> # GROUP 1, IDENTITY 7: U_A5(p0*t^(-6))
   EP5_1_7 := normal(A5*p0/t^6) :
   RHS5_1_7 := expand(p0*(7^13*t^7 + 54*7^11*t^6 + 1242*7^9*t^5 + 13110*7^7*t^4
       + 8947*7^6*t^3 + 2139*7^5*t^2-1541*7^2*t + 30360) + p1*(-7^12*t^6-53
       *7^10*t^5-1196*7^8*t^4-1748*7^7*t^3-8073*7^5*t^2-13018*7^3*t + 874
       *7-3795*t^(-1)) + 7^12*t^6 + 53*7^10*t^5 + 1196*7^8*t^4 + 1748*7^7*t^3
```

```
    + 8073 * 7^5 * t^2 + 13018 * 7^3 * t - 874 * 7 + 3795 * t^(-1)) :
  proof5_1_7 := provemodfuncGAMMA0UpETAidBATCH(EP5_1_7, 7, RHS5_1_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 23    This means checking up to q^(24).
We find that LHS - RHS is
```

$$O(q^{51})$$

$$proof5_1_7 := [1, -23, O(q^{51})] \tag{7}$$

```
> # GROUP 2, IDENTITY 1: U_A5(p1)
  EP5_2_1 := expand(A5 * p1) :
  RHS5_2_1 := expand(p0 * (-4 * 7^9 * t^6 - 12 * 7^8 * t^5 - 108 * 7^6 * t^4 - 480 * 7^4 * t^3
      - 984 * 7^2 * t^2 - 88 * 7 * t) + p1 * (7^10 * t^6 + 22 * 7^8 * t^5 + 29 * 7^7 * t^4 + 152 * 7
      ^5 * t^3 + 492 * 7^3 * t^2 + 884 * 7 * t + 11 * 7) - 5 * 7^8 * t^5 - 99 * 7^6 * t^4 - 809 * 7^4
      * t^3 - 3254 * 7^2 * t^2 - 883 * 7 * t - 11 * 7) :
  proof5_2_1 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_1, 7, RHS5_2_1, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$O(q^{48})$$

$$proof5_2_1 := [1, -20, O(q^{48})] \tag{8}$$

```
> # GROUP 2, IDENTITY 2: U_A5(p1*t^(-1))
  EP5_2_2 := expand(A5 * p1 / t) :
  RHS5_2_2 := 7 * t :
  proof5_2_2 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_2, 7, RHS5_2_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
```

```
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 1    This means checking up to q^(2).
We find that LHS - RHS is
```

$$O(q^{29})$$

$$proof5_2_2 := [1, -1, O(q^{29})] \quad \textbf{(9)}$$

```
> # GROUP 2, IDENTITY 3: U_A5(p1*t^(-2))
  EP5_2_3 := expand(A5*p1/t^2):
  RHS5_2_3 := expand(p0*(-8*7^3*t^2-64*7*t-8)+p1*(8*7^2*t+57+t^(-1))
      -7^3*t^2-58*7*t-57-t^(-1))):
  proof5_2_3 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_3, 7, RHS5_2_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof5_2_3 := [1, -9, O(q^{37})] \quad \textbf{(10)}$$

```
> # GROUP 2, IDENTITY 4: U_A5(p1*t^(-3))
  EP5_2_4 := expand(A5*p1/t^3):
  RHS5_2_4 := expand(p0*(4*7^5*t^3+92*7^3*t^2+540*7*t+80)+p1*(-4*7^4
      *t^2-89*7^2*t-482-10*t^(-1))+7^5*t^3+38*7^3*t^2+633*7*t+482
      +10*t^(-1))):
  proof5_2_4 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_4, 7, RHS5_2_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 12    This means checking up to q^(13).
```

We find that LHS - RHS is

$$O(q^{40})$$

$$proof5_2_4 := [1, -12, O(q^{40})] \quad \textbf{(11)}$$

```
> # GROUP 2, IDENTITY 5: U_A5(p1*t^(-4))
  EP5_2_5 := expand(A5*p1/t^4):
  RHS5_2_5 := expand(p0*(-20*7^5*t^3-484*7^3*t^2-436*7^2*t-632)+p1*(3
      *7^5*t^2+472*7^2*t+8*7^3+79*t^(-1)))-7^7*t^4-18*7^5*t^3-237*7
      ^3*t^2-480*7^2*t-8*7^3-79*t^(-1)):
  proof5_2_5 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_5, 7, RHS5_2_5, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 14    This means checking up to q^(15).
```

We find that LHS - RHS is

$$O(q^{42})$$

$$proof5_2_5 := [1, -14, O(q^{42})] \quad \textbf{(12)}$$

```
> # GROUP 2, IDENTITY 6: U_A5(p1*t^(-5))
  EP5_2_6 := expand(A5*p1/t^5):
  RHS5_2_6 := expand(p0*(8*7^7*t^4+20*7^6*t^3+292*7^4*t^2+1964*7^2*t
      +4360)+p1*(-7^7*t^3-20*7^5*t^2-286*7^3*t-1790*7-545*t^(-1)))-7
      ^9*t^5+25*7^6*t^3+230*7^4*t^2+2046*7^2*t+1790*7+545*t^(-1)):
  proof5_2_6 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_6, 7, RHS5_2_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
```

We find that LHS - RHS is

$$O(q^{45})$$

$$proof5_2_6 := [1, -17, O(q^{45})] \quad \textbf{(13)}$$

```
> # GROUP 2, IDENTITY 7: U_A5(p1*t^(-6))
  EP5_2_7 := expand(A5*p1/t^6):
  RHS5_2_7 := expand(p0*(-4*7^11*t^6-116*7^9*t^5-1460*7^7*t^4-176*7^7*t
      ^3-3560*7^4*t^2-7992*7^2*t-26624)+p1*(4*7^10*t^5+16*7^9*t^4
      +194*7^7*t^3+1123*7^5*t^2+3226*7^3*t+7425*7+3328*t^(-1))-11
      *7^9*t^5-680*7^7*t^4-1433*7^6*t^3-8391*7^4*t^2-22809*7^2*t-51974
      -3328*t^(-1)):
  proof5_2_7 := provemodfuncGAMMA0UpETAidBATCH(EP5_2_7, 7, RHS5_2_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 20    This means checking up to q^(21).
We find that LHS - RHS is
```

$$O(q^{48})$$

$$proof5_2_7 := [1, -20, O(q^{48})] \quad \textbf{(14)}$$

```
> # GROUP 3, IDENTITY 1: U_A5(1)
  EP5_3_1 := normal(A5):
  RHS5_3_1 := expand(p0*(4*7^10*t^6+92*7^8*t^5+828*7^6*t^4+3552*7^4*t^3
      +992*7^3*t^2+632*7*t)+p1*(-4*7^9*t^5-89*7^7*t^4-776*7^5*t^3
      -3231*7^3*t^2-880*7^2*t-79*7)+7^10*t^6+45*7^8*t^5+727*7^6*t^4
      +5687*7^4*t^3+22807*7^2*t^2+6161*7*t+79*7):
  proof5_3_1 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_1, 7, RHS5_3_1, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof5_3_1 := [1, -19, O(q^{47})] \quad \textbf{(15)}$$

```
> # GROUP 3, IDENTITY 2: U_A5(t^(-1))
```

```
EP5_3_2 := normal(A5/t) :
RHS5_3_2 := expand(p0*(8*7*t+8)+p1*(-8-t^(-1))+7*t+8+t^(-1)) :
proof5_3_2 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_2, 7, RHS5_3_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$O(q^{34})$$

$$proof5_3_2 := [1, -6, O(q^{34})] \quad \textbf{(16)}$$

```
> # GROUP 3, IDENTITY 3: U_A5(t^(-2))
EP5_3_3 := normal(A5/t^2) :
RHS5_3_3 := expand(p0*(-8*7*t-24)+p1*(10+3*t^(-1))-7^3*t^2-2*7*t
    -10-3*t^(-1)) :
proof5_3_3 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_3, 7, RHS5_3_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 8    This means checking up to q^(9).
We find that LHS - RHS is
```

$$O(q^{36})$$

$$proof5_3_3 := [1, -8, O(q^{36})] \quad \textbf{(17)}$$

```
> # GROUP 3, IDENTITY 4: U_A5(t^(-3))
EP5_3_4 := normal(A5/t^3) :
RHS5_3_4 := expand(p0*(4*7^5*t^3+4*7^4*t^2-4*7^2*t+80)+p1*(-4*7^4*t
    ^2-24*7^2*t+2*7-10*t^(-1))+7^5*t^3+38*7^3*t^2+178*7*t-2*7
    +10*t^(-1)) :
proof5_3_4 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_4, 7, RHS5_3_4, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
```

```
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 12    This means checking up to q^(13).
We find that LHS - RHS is
```

$$O(q^{40})$$

$$proof5_3_4 := [1, -12, O(q^{40})] \quad \textbf{(18)}$$

```
> # GROUP 3, IDENTITY 5: U_A5(t^(-4))
  EP5_3_5 := normal(A5/t^4):
  RHS5_3_5 := expand(p0*(-8*7^7*t^4-172*7^5*t^3-20*7^5*t^2-12*7^3*t-88)
      + p1*(8*7^6*t^3+164*7^4*t^2+124*7^3*t+76*7+11*t^(-1))-7^7*t^4
      -74*7^5*t^3-1238*7^3*t^2-876*7^2*t-76*7-11*t^(-1)):
  proof5_3_5 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_5, 7, RHS5_3_5, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof5_3_5 := [1, -15, O(q^{43})] \quad \textbf{(19)}$$

```
> # GROUP 3, IDENTITY 6: U_A5(t^(-5))
  EP5_3_6 := normal(A5/t^5):
  RHS5_3_6 := expand(p0*(-8*7^9*t^5-48*7^7*t^4+20*7^7*t^3+1196*7^4*t^2
      +1044*7^2*t-2024)+p1*(8*7^8*t^4+6*7^7*t^3-136*7^5*t^2-1062*7^3
      *t-886*7+253*t^(-1))-7^9*t^5-8*7^8*t^4-24*7^6*t^3+1042*7^4*t^2
      +7478*7^2*t+886*7-253*t^(-1)):
  proof5_3_6 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_6, 7, RHS5_3_6, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
```

```
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 18    This means checking up to q^(19).
We find that LHS - RHS is
```

$$O(q^{46})$$

$$proof5_3_6 := [1, -18, O(q^{46})] \quad \textbf{(20)}$$

```
> # GROUP 3, IDENTITY 7: U_A5(t^(-6))
  EP5_3_7 := normal(A5/t^6) :
  RHS5_3_7 := expand(p0*(4*7^11*t^6+260*7^9*t^5+452*7^8*t^4+1312*7^6*t
      ^3-3256*7^4*t^2-4664*7^2*t+3952*7)+p1*(-4*7^10*t^5-257*7^8*t^4
      -426*7^7*t^3-1157*7^5*t^2+2882*7^3*t+3655*7-494*7*t^(-1))+45
      *7^9*t^5+1903*7^7*t^4+2907*7^6*t^3+7569*7^4*t^2-20401*7^2*t
      -25584+494*7*t^(-1))) :
  proof5_3_7 := provemodfuncGAMMA0UpETAidBATCH(EP5_3_7, 7, RHS5_3_7, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 21    This means checking up to q^(22).
We find that LHS - RHS is
```

$$O(q^{49})$$

$$proof5_3_7 := [1, -21, O(q^{49})] \quad \textbf{(21)}$$

```
> # GROUP 4, IDENTITY 1: U_B5(p0)
  EP5_4_1 := normal(B5*p0) :
  RHS5_4_1 := expand(p0*(8*7^7*t^4+80*7^5*t^3+216*7^3*t^2+79*7*t)+p1
      *(7^10*t^5+19*7^8*t^4+18*7^7*t^3+327*7^4*t^2+34*7^3*t+4)+7^9
      *t^5+11*7^7*t^4+38*7^5*t^3+31*7^3*t^2+6*7*t) :
  proof5_4_1 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_1, 7, RHS5_4_1, 14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
```

```
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof5_4_1 := [1, -15, O(q^{43})] \quad \textbf{(22)}$$

```
> # GROUP 4, IDENTITY 2: U_B5(p0*t^(-1))
  EP5_4_2 := normal(B5*p0/t):
  RHS5_4_2 := expand(p0*(-7^3*t^2-8*7*t)+p1*(7^2*t+7)+1-7^2*t):
  proof5_4_2 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_2, 7, RHS5_4_2, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$O(q^{34})$$

$$proof5_4_2 := [1, -6, O(q^{34})] \quad \textbf{(23)}$$

```
> # GROUP 4, IDENTITY 3: U_B5(p0*t^(-2))
  EP5_4_3 := normal(B5*p0/t^2):
  RHS5_4_3 := expand(p0*(-7^5*t^3-8*7^3*t^2)+p1*(7^4*t^2+7^3*t)-7^4*t^2
      -7^3*t-12):
  proof5_4_3 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_3, 7, RHS5_4_3, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof5_4_3 := [1, -9, O(q^{37})] \quad \textbf{(24)}$$

```
> # GROUP 4, IDENTITY 4: U_B5(p0*t^(-3))
  EP5_4_4 := normal(B5*p0/t^3):
```

```
RHS5_4_4 := expand(p0*(1+6*7^7*t^4+62*7^5*t^3+27*7^4*t^2+10*7^2*t)
    +p1*(-6*7^6*t^3-8*7^5*t^2-3*7^4*t-4*7)+2*7^6*t^3+24*7^4*t^2
    +83*7^2*t+108):
proof5_4_4 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_4,7,RHS5_4_4,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 13    This means checking up to q^(14).
We find that LHS - RHS is
```

$$O(q^{41})$$

$$proof5_4_4 := [1, -13, O(q^{41})] \quad \textbf{(25)}$$

```
> # GROUP 4, IDENTITY 5: U_B5(p0*t^(-4))
EP5_4_5 := normal(B5*p0/t^4):
RHS5_4_5 := expand(p0*(-7^9*t^5-50*7^7*t^4-60*7^6*t^3-162*7^4*t^2-60
    *7^2*t-6)+p1*(7^8*t^4+7^8*t^3+54*7^5*t^2+18*7^4*t+24*7)-7^8
    *t^4-25*7^6*t^3-186*7^4*t^2-498*7^2*t-120*7):
proof5_4_5 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_5,7,RHS5_4_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$O(q^{44})$$

$$proof5_4_5 := [1, -16, O(q^{44})] \quad \textbf{(26)}$$

```
> # GROUP 4, IDENTITY 6: U_B5(p0*t^(-5))
EP5_4_6 := normal(B5*p0/t^5):
RHS5_4_6 := expand(p0*(-7^11*t^6-8*7^9*t^5+3*7^9*t^4+30*7^7*t^3+81
    *7^5*t^2+30*7^3*t+3*7)+p1*(7^10*t^5+7^9*t^4-3*7^8*t^3-27*7^6
    *t^2-9*7^5*t-12*7^2)-7^10*t^5-7^9*t^4+9*7^7*t^3+93*7^5*t^2
    +249*7^3*t+836*7):
```

```
proof5_4_6 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_6, 7, RHS5_4_6, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof5_4_6 := [1, -19, O(q^{47})] \quad \textbf{(27)}$$

```
> # GROUP 4, IDENTITY 7: U_B5(p0*t^(-6))
  EP5_4_7 := normal(B5*p0/t^6):
  RHS5_4_7 := expand(p0*(-7^13*t^7-8*7^11*t^6+2*7^9*t^4+20*7^7*t^3+54
      *7^5*t^2+20*7^3*t+2*7)+p1*(7^12*t^6+7^11*t^5-2*7^8*t^3-18*7
      ^6*t^2-6*7^5*t-8*7^2)-7^12*t^6-7^11*t^5+6*7^7*t^3+62*7^5*t^2
      +166*7^3*t-748*7^2):
  proof5_4_7 := provemodfuncGAMMA0UpETAidBATCH(EP5_4_7, 7, RHS5_4_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 22    This means checking up to q^(23).
We find that LHS - RHS is
```

$$O(q^{50})$$

$$proof5_4_7 := [1, -22, O(q^{50})] \quad \textbf{(28)}$$

```
> # GROUP 5, IDENTITY 1: U_B5(p1*t^(-1))
  EP5_5_1 := expand(B5*p1/t):
  RHS5_5_1 := expand(p0*(8*7^8*t^4+80*7^6*t^3+216*7^4*t^2+552*7*t)+p1
      *(7^11*t^5+19*7^9*t^4+18*7^8*t^3+327*7^5*t^2+34*7^4*t+29)-1
      +7^10*t^5+11*7^8*t^4+38*7^6*t^3+31*7^4*t^2+41*7*t):
  proof5_5_1 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_1, 7, RHS5_5_1, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
```

```
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 15    This means checking up to q^(16).
We find that LHS - RHS is
```

$$O(q^{43})$$

$$proof5_5_1 := [1, -15, O(q^{43})] \quad \textbf{(29)}$$

```
> # GROUP 5, IDENTITY 2: U_B5(p1*t^(-2))
  EP5_5_2 := expand(B5*p1/t^2):
  RHS5_5_2 := expand(p0*(-8*7^3*t^2-8*7^2*t)+p1*(8*7^2*t+7^2)-7^3*t^2
      -8*7^2*t+11):
  proof5_5_2 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_2, 7, RHS5_5_2, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 7    This means checking up to q^(8).
We find that LHS - RHS is
```

$$O(q^{35})$$

$$proof5_5_2 := [1, -7, O(q^{35})] \quad \textbf{(30)}$$

```
> # GROUP 5, IDENTITY 3: U_B5(p1*t^(-3))
  EP5_5_3 := expand(B5*p1/t^3):
  RHS5_5_3 := expand(p0*(-8*7^5*t^3-8*7^4*t^2)+p1*(8*7^4*t^2+7^4*t)-7
      ^5*t^3-8*7^4*t^2-7^4*t-96):
  proof5_5_3 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_3, 7, RHS5_5_3, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 10    This means checking up to q^(11).
```

We find that LHS - RHS is

$$O(q^{38})$$

$$proof5_5_3 := [1, -10, O(q^{38})] \quad \textbf{(31)}$$

```
> # GROUP 5, IDENTITY 4: U_B5(p1*t^(-4))
  EP5_5_4 := expand(B5*p1/t^4):
  RHS5_5_4 := expand(p0*(48*7^7*t^4+72*7^6*t^3+216*7^4*t^2+80*7^2*t
      +8)+p1*(-48*7^6*t^3-65*7^5*t^2-24*7^4*t-32*7)-7^7*t^4+16*7^6
      *t^3+199*7^4*t^2+664*7^2*t+740):
  proof5_5_4 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_4,7,RHS5_5_4,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 14    This means checking up to q^(15).
```

We find that LHS - RHS is

$$O(q^{42})$$

$$proof5_5_4 := [1, -14, O(q^{42})] \quad \textbf{(32)}$$

```
> # GROUP 5, IDENTITY 5: U_B5(p1*t^(-5))
  EP5_5_5 := expand(B5*p1/t^5):
  RHS5_5_5 := expand(p0*(-8*7^9*t^5-64*7^8*t^4-80*7^7*t^3-216*7^5*t^2
      -80*7^3*t-8*7)+p1*(8*7^8*t^4+9*7^8*t^3+72*7^6*t^2+24*7^5*t
      +32*7^2)-7^9*t^5-8*7^8*t^4-31*7^7*t^3-248*7^5*t^2-664*7^3*t
      -740*7):
  proof5_5_5 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_5,7,RHS5_5_5,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 17    This means checking up to q^(18).
```

We find that LHS - RHS is

$$O(q^{45})$$

(33)

$$proof5_5_5 := [1, -17, O(q^{45})] \quad (33)$$

```
> # GROUP 5, IDENTITY 6: U_B5(p1*t^(-6))
  EP5_5_6 := expand(B5*p1/t^6):
  RHS5_5_6 := expand(p0*(-8*7^11*t^6-8*7^10*t^5+40*7^9*t^4+400*7^7*t^3
      +1080*7^5*t^2+400*7^3*t+40*7)+p1*(8*7^10*t^5+7^10*t^4-40*7^8
      *t^3-360*7^6*t^2-120*7^5*t-160*7^2)-7^11*t^6-8*7^10*t^5-7^10*t^4
      +120*7^7*t^3+1240*7^5*t^2+3320*7^3*t+4720*7):
  proof5_5_6 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_6,7,RHS5_5_6,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof5_5_6 := [1, -19, O(q^{47})] \quad (34)$$

```
> # GROUP 5, IDENTITY 7: U_B5(p1*t^(-7))
  EP5_5_7 := expand(B5*p1/t^7):
  RHS5_5_7 := expand(p0*(-8*7^13*t^7-8*7^12*t^6-136*7^9*t^4-1360*7^7*t^3
      -3672*7^5*t^2-1360*7^3*t-136*7)+p1*(8*7^12*t^6+6*7^11*t^5-19*7
      ^9*t^4+818*7^7*t^3+1167*7^6*t^2+2798*7^4*t+11*7^4+t^(-1))+6
      *7^13*t^7+20*7^12*t^6+272*7^10*t^5+1893*7^8*t^4+435*7^7*t^3
      -2953*7^5*t^2-10620*7^3*t-27029*7):
  proof5_5_7 := provemodfuncGAMMA0UpETAidBATCH(EP5_5_7,7,RHS5_5_7,14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 24    This means checking up to q^(25).
We find that LHS - RHS is
```

$$O(q^{52})$$

$$proof5_5_7 := [1, -24, O(q^{52})] \quad (35)$$

```
> # GROUP 6, IDENTITY 1: U_B5(1) = U_7(1) = 1, directly from the definition of U_7.

> # GROUP 6, IDENTITY 2: U_B5(t^(-1))
  EP5_6_2 := normal(B5/t):
  RHS5_6_2 := -7*t-4:
  proof5_6_2 := provemodfuncGAMMA0UpETAidBATCH(EP5_6_2, 7, RHS5_6_2, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 3    This means checking up to q^(4).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{31})$$

$$proof5_6_2 := [1, -3, \mathrm{O}(q^{31})] \qquad \textbf{(36)}$$

```
> # GROUP 6, IDENTITY 3: U_B5(t^(-2))
  EP5_6_3 := normal(B5/t^2):
  RHS5_6_3 := -7^3*t^2+20:
  proof5_6_3 := provemodfuncGAMMA0UpETAidBATCH(EP5_6_3, 7, RHS5_6_3, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 6    This means checking up to q^(7).
We find that LHS - RHS is
```

$$\mathrm{O}(q^{34})$$

$$proof5_6_3 := [1, -6, \mathrm{O}(q^{34})] \qquad \textbf{(37)}$$

```
> # GROUP 6, IDENTITY 4: U_B5(t^(-3))
  EP5_6_4 := normal(B5/t^3):
  RHS5_6_4 := -7^5*t^3-88:
  proof5_6_4 := provemodfuncGAMMA0UpETAidBATCH(EP5_6_4, 7, RHS5_6_4, 14);

*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
```

```
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 9    This means checking up to q^(10).
We find that LHS - RHS is
```

$$O(q^{37})$$

$$proof5_6_4 := [1, -9, O(q^{37})] \quad \textbf{(38)}$$

```
> # GROUP 6, IDENTITY 5: U_B5(t^(-4))
  EP5_6_5 := normal(B5/t^4):
  RHS5_6_5 := -7^7*t^4+260:
  proof5_6_5 := provemodfuncGAMMA0UpETAidBATCH(EP5_6_5,7,RHS5_6_5,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 13    This means checking up to q^(14).
We find that LHS - RHS is
```

$$O(q^{41})$$

$$proof5_6_5 := [1, -13, O(q^{41})] \quad \textbf{(39)}$$

```
> # GROUP 6, IDENTITY 6: U_B5(t^(-5))
  EP5_6_6 := normal(B5/t^5):
  RHS5_6_6 := -7^9*t^5+68*7:
  proof5_6_6 := provemodfuncGAMMA0UpETAidBATCH(EP5_6_6,7,RHS5_6_6,14);
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 16    This means checking up to q^(17).
We find that LHS - RHS is
```

$$O(q^{44})$$

(40)

$$proof5_6_6 := [1, -16, O(q^{44})] \quad \textbf{(40)}$$

```
> # GROUP 6, IDENTITY 7: U_B5(t^(-6))
  EP5_6_7 := normal(B5/t^6):
  RHS5_6_7 := -7^11*t^6-2392*7:
  proof5_6_7 := provemodfuncGAMMA0UpETAidBATCH(EP5_6_7, 7, RHS5_6_7, 14);
```

```
*** There were NO errors.
*** o each EP in gcombo is an MF on Gamma[0](98)
*** o Each term in the etacombo is a  modular function on
      Gamma0(14).
*** o We also checked that the total order of
      each term etacombo was zero.
*** To prove the identity U[7](EP)=etacombo we need to show
    that v[oo](ID) > 19    This means checking up to q^(20).
We find that LHS - RHS is
```

$$O(q^{47})$$

$$proof5_6_7 := [1, -19, O(q^{47})] \quad \textbf{(41)}$$

```
>
```